Building an Investment Trading Strategy – Quick Reference Sheet

1. Hypothesis Development

- Hypothesis: Periods exist when either SPY or RSP outperforms based on market conditions.
- SPY: Market-cap weighted; favors large-cap stocks.
- RSP: Equal-weighted; favors small- and mid-cap stocks.
- Objective: Identify signals to determine which ETF to invest in.

2. Exploratory Data Analysis (EDA)

- Retrieve historical prices (e.g., from Yahoo Finance).
- Analyze cumulative returns to observe long-term trends.
- Calculate rolling returns to capture performance over different overlapping periods.

3. Rolling Returns

- Definition: Rolling returns are calculated over a specified time window (e.g., 1 month, 6 months, 1 year).
- Purpose: Identify consistency and performance trends across various market conditions.
- Key Metrics: Percentage of periods RSP outperforms SPY, maximum return, minimum return.

4. Z-Score Analysis

- Z-Score: Measures the relative performance of SPY vs. RSP in terms of standard deviations from the mean.
- Expanding Z-Scores: Prevent look-ahead bias by using only past data.
- Use Cases:
  - Mean-reversion: Switch when z-scores exceed a threshold.
  - Momentum: Identify sustained performance trends.

5. Train-Test Split

- Split the dataset into:
  - In-Sample (Training): Historical data for parameter optimization.
  - Out-of-Sample (Testing): New data for validating the strategy's robustness.
- Avoid look-ahead bias: Perform normalization and standardization independently on train and test sets.

6. Volatility Overlay

- Purpose: Mitigate extreme downside risk and enhance stability.
- Method:
  - High volatility threshold: Move to cash.
  - Low volatility threshold: Re-enter the market.

- Benefits: Preserve capital, improve risk-adjusted returns, and adapt to changing market conditions.

7. Backtesting

- Process:
  - Evaluate positions based on z-scores and thresholds.
  - Use daily returns to calculate strategy performance.
  - Incorporate a volatility filter for enhanced risk management.
- Metrics:
  - Strategy returns vs. benchmark (SPY).
  - Cumulative returns over time.

8. Performance Metrics

- Rolling Sharpe Ratio: Evaluate risk-adjusted returns over a rolling window.
- Rolling Maximum Drawdown: Assess the largest declines in portfolio value within a specified window.
- Objective: Understand the trade-offs between return and risk.

9. Out-of-Sample Testing

- Purpose: Validate strategy robustness on unseen data.
- Use the last position from in-sample testing as the starting point.
- Compare out-of-sample performance to benchmark returns.